

D5 – DESIGN RATIONALE DOCUMENT	
Subject	ITA301 – Information System Analysis & Design
Deliverable	D5 – Final Presentation Support Document
Group	Group 3 AgriUAV Platform FPT University HCM Summer 2026
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This document records 10 major architectural and technical design decisions made during D4 (System Design), with alternatives considered and engineering/field-level justification. Each decision is linked to the FR/NFR/UC it satisfies, enabling cross-referencing during oral viva.

#	Decision (What)	Alternative Considered	Justification (Why / Engineering Rationale)	FR / NFR / UC
1	Layered Modular Monolith Architecture	Distributed Microservices Architecture	Ensures immediate database-level transaction consistency (Strong ACID) to eliminate race conditions and double-booking of drone resources (UC02). Drastically minimises operational complexity — HTX cooperatives have no dedicated DevOps team (D4 §4.2). Single codebase enables hotfix deployment within ≤ 4 hours (NFR07).	NFR07, NFR04, UC02
2	Redis In-Memory Buffer Layer	Direct-to-Disk Database Ingestion	At $4\text{Hz} \times 50$ active drones = 200 req/s (NFR04 peak baseline), direct PostgreSQL writes cause I/O disk bottlenecks. Redis acts as a high-performance "shock absorber" absorbing telemetry streams to RAM; a background worker asynchronously batch-flushes to TimescaleDB. Prevents database contention while maintaining $\leq 500\text{ms}$ latency (NFR01).	NFR01, NFR04, UC04
3	PostGIS Spatial Extension (GEOMETRY type)	Flat Geographic Coordinate Strings (TEXT type)	Enables the database engine to natively execute geometric evaluations (ST_Contains, ST_Distance, ST_GeomFromGeoJSON) for field boundary checks and geofence proximity alerts. Critical for FR12 (50m buffer around 110kV	FR01, FR12, UC09, NFR01

			lines) and UC09 (polygon area auto-computation via ST_Area). TEXT columns cannot support these spatial queries without full-table application-layer parsing.	
4	UUID v4 Primary Keys (VARCHAR 36)	Auto-Incrementing Integers (INT AUTO_INCREMENT)	With an Offline-First model, pilots generate flight log and check-in records in remote fields without server connectivity. UUID v4 guarantees global uniqueness without a server round-trip, preventing primary key collisions upon synchronisation (NFR06, ≤30s sync). INT sequences generated offline across multiple devices will collide on upload.	NFR06, FR09, FR13
5	TimescaleDB Time-Series Extension	Standard Native PostgreSQL Relational Engine	drone_positions and audit_logs accumulate high-frequency time-stamped rows (up to 200 inserts/s). TimescaleDB partitions these into automated hypertable chunks (partitioned weekly), preserving O(log N) query speed for time-range scans (e.g., export all GPS trail for a given flight). Standard PostgreSQL degrades to O(N) full-table scans on large telemetry tables.	NFR01, NFR04, FR13, FR10
6	Flutter with Encrypted Local SQLite (Offline-First)	Progressive Web App (PWA) / React Native	Flutter compiles to native ARM code, providing better rendering performance on low-spec Android devices common among DBSCL pilots (NFR03). Encrypted SQLite embedded in the Flutter client provides a resilient offline cache for check-in/out records when 4G drops (NFR06). React Native's JS bridge adds latency; PWA cannot access native GPS APIs reliably at ≤5m accuracy (FR09).	NFR03, NFR06, FR09, FR10
7	Leaflet.js with OpenStreetMap	Google Maps Platform API	Provides full GIS capabilities — polygon	FR01, FR10, FR12

			drawing for field boundaries (FR01), drone GPS trail overlays (FR10), geofence visualisation (FR12) — without per-request API licensing fees. Critical for cooperative budget alignment (D4 §5). Leaflet + OpenStreetMap is fully open-source; Google Maps charges per map load and per API call, creating unpredictable cost at 50-drone scale.	
8	Column-Level AES-256 Encryption at Rest	Disk-Level Encryption Only	Sensitive PII columns (phone numbers, GPS trails, land deed boundaries) are encrypted at the PostgreSQL column level using transparent AES-256 ciphers. Disk-level encryption protects only against physical theft of storage media; it does not prevent an application-layer SQL injection from reading plaintext PII in RAM. Column-level encryption satisfies NFR05's 'sensitive data at rest must be encrypted' outcome.	NFR05, FR18
9	JWT Signed via RSA-256 Asymmetric Keys	Traditional Server-Side Session Cookies	Decouples stateful session validation from the backend. Pilots' mobile apps can maintain cryptographic auth state offline and re-authenticate upon network reconnection without re-issuing credentials. Server-side session cookies require a database lookup on every request, adding latency and creating a single-point-of-failure for session storage. Satisfies NFR05 ('stateless token-based mechanisms with refresh capability').	NFR05, NFR06, FR18
10	Hardware-Bound (Device ID) Rate Limiting	Traditional IP-Based Rate Limiting	Protects the telemetry ingestion gateway from malfunctioning drone-side sensors flooding the endpoint with corrupt packets (DoS)	NFR04, FR21, §7.1 STRIDE

			scenario per STRIDE analysis). IP-based limiting would block all pilots sharing the same 4G tower NAT IP. Device ID binding (drone serial number from FR21) drops only the specific malfunctioning hardware ID, preserving service for other active missions.	
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